

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [5]: df = sns.load_dataset("titanic")
df.head()
```

```
Out[5]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	ad
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

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```
In [7]: df.describe()
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Out[7]:
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   survived        891 non-null    int64
 1   pclass          891 non-null    int64
 2   sex             891 non-null    object
 3   age             714 non-null    float64
 4   sibsp           891 non-null    int64
 5   parch           891 non-null    int64
 6   fare            891 non-null    float64
 7   embarked        889 non-null    object
 8   class           891 non-null    category
 9   who             891 non-null    object
10   adult_male      891 non-null    bool
11   deck            203 non-null    category
12   embark_town     889 non-null    object
13   alive           891 non-null    object
14   alone           891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

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In [13]: df.isnull().sum()
```

```
Out[13]: survived        0
pclass          0
sex             0
age             177
sibsp           0
parch           0
fare            0
embarked        2
class           0
who             0
adult_male      0
deck            688
embark_town     2
alive           0
alone           0
dtype: int64
```

```
In [17]: df['age'] = df['age'].fillna(np.mean(df['age']))
df['deck'] = df['deck'].fillna(df['deck'].mode()[0])
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
```

```
In [21]: df.isnull().sum()
```

```
Out[21]: survived      0
         pclass        0
         sex           0
         age           0
         sibsp         0
         parch         0
         fare          0
         embarked      0
         class         0
         who           0
         adult_male     0
         deck          0
         embark_town    2
         alive         0
         alone         0
         dtype: int64
```

```
In [25]: import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(
    x='sex',
    y='age',
    hue='survived',
    data=df,
    palette='Set2'
)

plt.title('Distribution of Age by Gender and Survival Status')
plt.show()
```

